

Brief Introduction to Python

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June 16, 2025

1 Variables

In Python, variables are dynamically typed. A variable can hold values of any type, and its type may change during execution.

```
1 x = 5 # Integer
2 name = "Alice" # String
3 pi = 3.1415 # Float
4 is_valid = True # Boolean
5
```

Exercises

1. Declare a variable 'age' with your age and print its type using the 'type' function.
2. Assign a string to a variable 'greeting' and concatenate it with a name variable.
3. Change the type of 'x' from integer to string and print both values.

2 Lists, Dictionaries and Sets

2.1 Lists

A list is an ordered and mutable collection of elements. Elements can be of mixed types.

```
1 fruits = ["apple", "banana", "cherry"]
2 print(fruits[1]) # Output: banana
3 fruits.append("date") # Add an element
4 fruits[0] = "avocado" # Modify an element
5
```

Exercises

1. Create a list of five numbers and compute its sum.
2. Reverse a list 'colors = ["red", "green", "blue"]' and print the result.
3. Remove the last element from a list and print the remaining elements.

2.2 Dictionaries

A dictionary stores data as key-value pairs. Keys must be unique and immutable.

```
1 person = {"name": "Alice", "age": 25}
2 print(person["name"]) # Output: Alice
3 person["city"] = "Rome" # Add new key-value pair
4
```

Exercises

1. Create a dictionary mapping three countries to their capitals.
2. Update the dictionary by changing one capital and print the changes.
3. Write a loop to print all keys and values of the dictionary.

2.3 Sets

A set is an unordered collection of unique elements. Useful for membership tests and eliminating duplicates.

```
1 unique_numbers = {1, 2, 3, 2}
2 print(unique_numbers)      # Output: {1, 2, 3}
3 unique_numbers.add(4)      # Add an element
4
```

Exercises

1. Create two sets and compute their union, intersection and difference.
2. Remove a specific element from a set.
3. Check if one set is a subset of another.

3 List Comprehension

List comprehensions provide a concise syntax to create lists based on existing iterables.

```
1 squares = [x**2 for x in range(5)]
2 print(squares)      # Output: [0, 1, 4, 9, 16]
3 evencubes = [x**3 for x in range(10) if x % 2 == 0]
4 print(even_cubes)    # Output: [0, 8, 64, 216, 512]
5
```

Exercises

1. Use a list comprehension to extract the first letter of each word in 'words = ["hello", "world"]'.
2. Create a list of tuples '(x, x*2)' for 'x' in range 5.
3. Flatten a nested list '[[1,2],[3,4],[5]]' using a single comprehension.

4 Functions

Functions allow organization of code into reusable blocks. Define a function with 'def', include parameters and return values.

```
1 def greet(name):
2     """Return a greeting for the given name."""
3     return f"Hello, {name}!"
4
5 print(greet("Alice"))  # Output: Hello, Alice!
6
7 # Function with default parameter
8
9 def power(base, exponent=2):
10     """Return base raised to the given exponent."""
```

```

11     return base ** exponent
12
13     print(power(3))           # Output: 9
14     print(power(2, 3))       # Output: 8
15
16     # Function with arbitrary arguments
17
18     def summarize(*args):
19         """Return sum of all arguments."""
20         return sum(args)
21
22     print(summarize(1, 2, 3, 4)) # Output: 10
23

```

Exercises

1. Write a function ‘square(x)’ that returns the square of x.
2. Create a function that takes a list and returns its maximum value.
3. Define a function that accepts keyword arguments and prints them.

5 NumPy

NumPy is a library for efficient numerical computing with support for large, multi-dimensional arrays and matrices.

```

1     import numpy as np
2
3     array = np.array([1, 2, 3])
4     print(array * 2)           # Output: [2 4 6]
5
6     matrix = np.array([[1, 2], [3, 4]])
7     print(matrix.T)           # Transpose
8     print(np.dot(matrix, matrix)) # Matrix multiplication
9

```

Exercises

1. Create a 3x3 identity matrix using NumPy.
2. Compute the mean and standard deviation of ‘array = np.arange(10)’.
3. Generate an array of 20 linearly spaced points between 0 and 1.

6 Matplotlib

Matplotlib is a plotting library for Python. It provides an object-oriented API and state-based interface.

```

1     import matplotlib.pyplot as plt
2     import numpy as np
3
4     x = np.linspace(0, 2* np.pi, 100)
5     y = np.sin(x)
6
7     plt.figure()
8     plt.plot(x, y)
9     plt.xlabel('x')

```

```
10 plt.ylabel('sin(x)')
11 plt.title('Sine Wave')
12 plt.grid(True)
13 plt.show()
14
```

Exercises

1. Plot a cosine function on the same axes as the sine function.
2. Create a scatter plot for random points using 'np.random.rand(50,2)'.
3. Save a plot to file 'output.png' without displaying it interactively.